

05.27.26



COS221

PRACTICAL ASSIGNMENT

5

Navigation aid

AGENDA

DOCUMENT WALKTHROUGH (8 MIN)

- ★ (E)ER Diagram
 - Key entities, cardinalities, design decisions, specialisation/generalisation, complex/multivalued attributes and assumptions
- ★ Relational Mapping
 - Walkthrough the steps with emphasis on M:N, weak entities and specialisation
- ★ Relational Schema
 - Visual diagram with PKs, FKs, constraints and data types are in SQL
- ★ Query Optimisation (Task 7)
 - Execution plan analysis, what was optimised and why performance changed
- ★ Data Sources (Task 6)
 - Where data came from and how database was populated

TRAVELLER INTERFACE (4 MIN)

- ★ Log in and show dedicated traveller interface
- ★ Browse destinations, flights, accommodations, attractions and restaurants
- ★ Compare travel packages from different agencies
- ★ Book a package and show the booking saved to the database
- ★ Leave a review and rating for an agency or package
- ★ Use the sort and filter options (price, destination, duration, rating, etc)
- ★ Open a detailed package view showing pricing, itinerary, images, agency info and reviews



AGENDA

AGENCY INTERFACE (4 MIN)

- ★ Log out and log in as a Travel Agency and show that it is a completely separate interface
- ★ Create a new travel package and add destinations, flights, accommodation, restaurants and attractions to it
- ★ Edit and delete an existing package
- ★ Create or manage a group trip offering

SECURITY AND TECHNICAL (2 MIN)

SQL injection prevention using prepared statements, example extracted from /tripistry/review_functions.php

```
71 $sql = "INSERT INTO review (traveller_id, agency_id, package_id, rating, comment, sentiment, created_date)
72     VALUES (:traveller_id, :agency_id, :package_id, :rating, :comment, :sentiment, NOW())";
73
74 $stmt = $db->prepare($sql);
75 $result = $stmt->execute([
76     ':traveller_id' => $travellerId,
77     ':agency_id' => $agencyId,
78     ':package_id' => $packageId,
79     ':rating' => $rating,
80     ':comment' => $comment,
81     ':sentiment' => $sentiment
82 ]);
83
```



AGENDA

SECURITY AND TECHNICAL (2 MIN)

Role based access control /tripistry/login.php

```
8  if (is_logged_in())
9  {
10     header('Location: ' . (current_role() === 'agency' ? BASE_URL . '/agency/dashboard.php' : BASE_URL . '/traveller/dashboard.php'));
11     exit;
12 }
13
```



Agency View

In /tripistry/agency/dashboard.php we define the agency view

```
login.php dashboard.php .../traveller dashboard.php .../agency
tripistry > agency > dashboard.php
Dian Le Roux, 2 days ago | 3 authors (ksmdragon-opensource and others)
1 <?php
2 /*agency/dashboard.php - Agency home page*/
3 require_once __DIR__ . '/../config/db.php';
4 require_once __DIR__ . '/../includes/auth.php';
5 require_role('agency');
6
7 $db = get_db();
8 $agency_id = (int)$SESSION['agency_id'];
9
10 // Stats
11 $stats = $db->prepare("
12     SELECT
13         (SELECT COUNT(*) FROM travel_package WHERE agency_id=?) AS pkg_count,
14         (SELECT COUNT(*) FROM group_trip WHERE agency_id=? AND status='open') AS open_trips,
15         (SELECT COUNT(*) FROM booking b JOIN travel_package tp ON tp.package_id=b.package_id WHERE tp.agency_id=?) AS total_bookings,
16         (SELECT ROUND(AVG(r.rating),1) FROM review r WHERE r.agency_id=?) AS avg_rating
17 ");
18 $stats->execute([$agency_id, $agency_id, $agency_id, $agency_id]);
19 $s = $stats->fetch();
20
21 // Recent bookings for this agency's packages
22 $stmt = $db->prepare("
23     SELECT b.booking_id, b.booking_date, b.total_price, b.status,
24           tp.name AS package_name,
25           t.first_name, t.last_name
26     FROM booking b
27     JOIN travel_package tp ON tp.package_id = b.package_id
28     JOIN traveller t ON t.traveller_id = b.traveller_id
29     WHERE tp.agency_id = ?
30     ORDER BY b.booking_date DESC
31     LIMIT 8
32 ");
33 $stmt->execute([$agency_id]);
34 $bookings = $stmt->fetchAll();
35
36 // Recent reviews
37 $stmt = $db->prepare("
38     SELECT r.*, tp.name AS package_name, t.first_name, t.last_name
39     FROM review r
40     LEFT JOIN travel_package tp ON tp.package_id = r.package_id
41     JOIN traveller t ON t.traveller_id = r.traveller_id
42     WHERE r.agency_id = ? OR r.package_id IN (SELECT package_id FROM travel_package WHERE agency_id=?)
43     ORDER BY r.created_date DESC
44     LIMIT 4
45 ");
46 $stmt->execute([$agency_id, $agency_id]);
47 $reviews = $stmt->fetchAll();
48
49 $display_name = $SESSION['display_name'] ?? 'Agency';
```

Traveller View

In /tripistry/traveller/dashboard.php we define the agency view

```
login.php dashboard.php .../traveller dashboard.php .../agency
tripistry > traveller > dashboard.php
ksmdragon-opensource, yesterday | 5 authors (ksmdragon-opensource and others)
1 <?php
2 /*Traveller home page*/
3 require_once __DIR__ . '/../config/db.php';
4 require_once __DIR__ . '/../includes/auth.php';
5 require_role('traveller');
6
7 $uid = current_user_id();
8 $db = get_db();
9
10 // Stats
11 $stmt = $db->prepare("SELECT COUNT(*) FROM booking WHERE traveller_id = ?"); $stmt->execute([$uid]);
12 $booking_count = $stmt->fetchColumn();
13
14 $stmt = $db->prepare("SELECT COUNT(*) FROM booking WHERE traveller_id = ? AND status = 'confirmed'"); $stmt->execute([$uid]);
15 $confirmed = $stmt->fetchColumn();
16
17 $stmt = $db->prepare("SELECT COUNT(*) FROM review WHERE traveller_id = ?"); $stmt->execute([$uid]);
18 $review_count = $stmt->fetchColumn();
19
20 // Recent bookings
21 $stmt = $db->prepare("
22     SELECT b.booking_id, b.booking_date, b.total_price, b.status,
23           tp.name AS package_name, ta.company_name AS agency_name
24     FROM booking b
25     JOIN travel_package tp ON tp.package_id = b.package_id
26     JOIN travel_agency ta ON ta.agency_id = tp.agency_id
27     WHERE b.traveller_id = ?
28     ORDER BY b.booking_date DESC
29     LIMIT 5");
30 $stmt->execute([$uid]);
31 $recent_bookings = $stmt->fetchAll();
32
33 // Recommended packages (simpleway of recommending, same destination as past bookings or top-rated)
34 $stmt = $db->prepare("
35     SELECT tp.package_id, tp.name, tp.base_price, tp.duration_days,
36           ta.company_name, tp.image_url,
37           ROUND(AVG(r.rating),1) AS avg_rating
38     FROM travel_package tp
39     JOIN travel_agency ta ON ta.agency_id = tp.agency_id
40     LEFT JOIN review r ON r.package_id = tp.package_id
41     WHERE tp.package_id NOT IN (
42         SELECT package_id FROM booking WHERE traveller_id = ?
43     )
44     GROUP BY tp.package_id
45     ORDER BY avg_rating DESC, tp.base_price ASC
46     LIMIT 4"); Marko de Swardt, last week • Added more comments for better understanding ...
47 $stmt->execute([$uid]);
48 $recommended = $stmt->fetchAll();
49
```



Q&A (5 MIN)

AGENDA

BONUS FEATURES (1-2 MIN)

- ★ AI integration
 - Review sentiments
 - Chatbot
- ★ Enhanced UI
 - Dark mode
 - Hand drawn icons/images
 - Extended interactivity
- ★ Hashing and encryption of passwords
- ★ CSRF
- ★ Cross-site scripting

